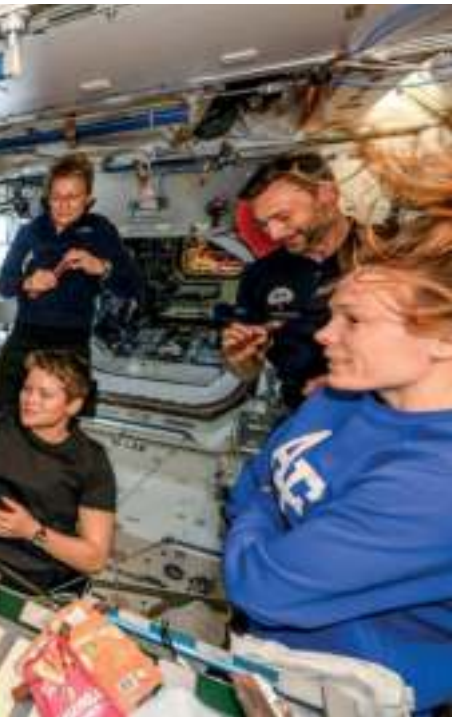




GLITTERING OVERVIEW Left, mealtime on the ISS; grab from a video shot by Shubhanshu showing India and Sri Lanka at night

> *"When I first saw India from space it was night and it was lit up like sparkling diamonds. You could clearly see the outline of the peninsula and make out cities like Mumbai, Delhi and Hyderabad. It still gives me goosebumps describing it; the pictures and videos just don't do it justice"*

Q. You have said you were not feeling too good during the first few days in space. Why was that?
That is a normal feeling astronauts get because there is no gravity. We are designed to live in gravity all our lives. So, when suddenly you are launched into an environment without gravity, your body rebels and resists adapting to this new situation. Some major things happen. Initially, your heart keeps pumping with the same force as on Earth, pushing more blood up. Then, all the fluid or blood in your legs, which was earlier flowing down because of gravity, starts going up to your head. So, your head starts becoming bigger. Not because you are an astronaut and travelling to a space station and feeling great but, physiologically. Your spine, as I said earlier, gets elongated. Everything in your stomach also starts floating. So, you do not feel hungry when you are on the station for the first few days. These are big changes and your body gradually gets used to it. But what is marvellous is that, within three or four days, that becomes the new normal. The body adjusts brilliantly and everything becomes perfectly alright.

Q. Are you a foodie? Did you like the food in space?
I am from Lucknow and food is something we are proud of. I do like food, but I've been very disciplined with what I eat. And when you're going up, you know it won't be the same because the food has to be contained in terms of size, weight and volume. Nutrition is important, so it is designed that way. But it's not bad. You do have a food tasting session on ground before you go, and you select your own food, what you like and what you don't. And then you can choose from the options available and those items are packed for you when you go up there.

Q. Does it involve eating a paste squeezed out of a tube?
No, that was the case earlier. Now, the food has to be in moist form, so that it sticks together and does not float around since surface tension is a primary force acting in space. For instance, you cannot have dry basmati rice, as it will explode and go everywhere. So, that is a challenge. We do have spoons, as moist foods attach to it. The water is in pouches, and we use straws to drink.

Q. Is it difficult to digest food in microgravity?
Fortunately, our digestive system works on peristalsis, or involuntary muscle contractions that push food down our tracts irrespective of gravity. That is how we are able to eat and digest food in space.

Q. How about taking a bath or shaving? Is that hard?
Yes, everything is, and you have to learn how to do things. For maintaining body hygiene, there are designated places. You cannot have a shower because falling water is difficult to catch. You have special hygiene towels instead, which are